

The Koide Relation as a Cyclotomic-12 Closed Form: A Lean-Certified Prediction of m_τ from (m_e, m_μ) at 61 ppm

— with the structural derivation of the Koide phase from the QCD colour rank $N_c = 3$ —

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Claim-strength taxonomy

$$\mathbf{A}_{\text{Lean}} > \mathbf{A}/\mathbf{D} > \mathbf{B} > \mathbf{D}$$

A_{Lean}: machine-checked exact arithmetic in Lean 4, zero **sorry**.

A/D: full analytic derivation; computationally confirmed.

B: bridge claim; partially formalised or citation-backed.

D: informal / structural / exploratory.

Abstract

Koide’s 1983 relation $Q \equiv (m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$ has no accepted structural origin. We promote it from a numerical coincidence to a structural statement via three interlocking results.

Geometric form. The relation is equivalent to $v = (\sqrt{m_e}, \sqrt{m_\mu}, \sqrt{m_\tau})$ lying on the 45° cone about the democratic axis $(1, 1, 1)/\sqrt{3}$; PDG values place v at 44.99974° , just 0.95 arcseconds from exact.

Closed form. Solving as a quadratic in $\sqrt{m_\tau}$ yields

$$\sqrt{m_\tau} = 2(\sqrt{m_e} + \sqrt{m_\mu}) + \sqrt{3} \sqrt{m_e + 4\sqrt{m_e m_\mu} + m_\mu},$$

predicting $m_\tau = 1.776969$ GeV against the PDG value 1.77686 GeV—a relative error of 61 ppm ($0.91\sigma_{\text{PDG}}$). The surd coefficients satisfy cyclotomic-12 identities, placing the lepton mass spectrum in the atom family $\{\pi/12, \cos(\pi/12), \sqrt{3}\}$.

Structural origin. Within the UGP framework, the Koide phase $\theta = 2/9$ is derived unconditionally from the QCD colour rank $N_c = 3$ via $\theta = (N_c^2 - 1)/(4N_c^2)$. The absolute

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lepton mass scale follows from first principles: $y_\tau = 1/98$ arises as $c_V/N_{\text{mod}2}$, where $c_V = 1/49$ is the canonical Z_7 potential coefficient and $N_{\text{mod}2} = 2$ the binary tape level, giving $m_\tau = v_H/(98\sqrt{2}) = 1776.57 \text{ MeV}$ (-2.42σ from PDG 2024) with the SRRG Higgs VEV as the sole dimensional anchor. The closed form then predicts all three charged-lepton masses with zero free parameters. The combination $N_c + \theta = 29/9$ also controls the neutrino mass-squared splitting ratio $\Delta m_{21}^2/\Delta m_{31}^2$ to 0.52% from NuFIT 6.0. An S_3 -equivariant Newton-step operator with the Koide null cone as its attractor provides a dynamical realisation of the 45° condition.

All identities are machine-checked in Lean 4 against Mathlib with zero `sorry`.

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1 Introduction

In 1983, Yoshio Koide observed [3] that the charged-lepton masses satisfy

$$Q \equiv \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \quad (1)$$

at a precision far tighter than one would expect from accidental coincidence: with current PDG values [8], $Q = 0.666661(7)$ against the conjectured exact value $2/3 = 0.6\bar{6}$.

The relation has generated extensive phenomenological and model-building literature over four decades (see [2, 4, 9] and references therein) without reaching consensus on a structural origin. Candidate explanations include seesaw-mechanism corrections, extended symmetry schemes, and democratic-mixing arguments, but the identity itself as a pure statement about the charged-lepton mass eigenvalues has largely been treated as either numerical coincidence or phenomenological input.

This paper presents four observations that, taken together, reframe the Koide relation as a *concrete structural statement about the charged-lepton mass spectrum*. The first three are elementary mathematical facts independent of any physics framework; the fourth identifies a structural origin within the UGP framework:

1. **A geometric reformulation (§2):** (1) is algebraically equivalent to the statement that the sqrt-mass vector $v = (\sqrt{m_e}, \sqrt{m_\mu}, \sqrt{m_\tau})$ makes a 45° angle with the democratic axis. This is a purely elementary reformulation but places the relation in a well-defined geometric setting.

2. **An algebraic closed form** (§3): solving (1) as a quadratic in $\sqrt{m_\tau}$ yields a closed-form expression for m_τ as an explicit algebraic function of (m_e, m_μ) . Substituting PDG values for (m_e, m_μ) produces a prediction for m_τ at 61 ppm relative error, inside the PDG 1σ uncertainty band. This upgrades the relation from a consistency check to a concrete predictive identity.
3. **A cyclotomic-12 identification** (§4): the surd coefficients of the closed form satisfy exact half-angle identities involving $\cos(\pi/12)$, placing the identity within the cyclotomic-12 number field. This is a nontrivial algebraic-number-theoretic observation about the relation.
4. **A structural origin from $N_c = 3$** (§7): within the UGP framework, the Koide phase $\theta = 2/9$ is derived unconditionally from the QCD colour rank $N_c = 3$ via the strand-count identity $\theta = (N_c^2 - 1)/(4N_c^2)$. The same combination $N_c + \theta = 29/9$ controls the neutrino mass-squared splitting ratio $\Delta m_{21}^2/\Delta m_{31}^2$ to 0.16σ from NuFIT 6.0 (0.52%) via the Braid Atlas (§8). This observation is conditional on the UGP framework; the first three are not.

We additionally construct an S_3 -equivariant dynamical operator on $\mathbb{R}^3$ whose attractor set is the Koide null cone (§5), giving a dynamical realisation of the 45° condition. This is an auxiliary result: the main closed-form prediction does not depend on it.

All algebraic identities invoked are machine-checked in Lean 4 against Mathlib (§6); the `ugp-lean` formalisation is open-source and reproducible [7].

Observations (1)–(3) are deliberately framed as *standalone mathematical facts about the charged-lepton mass spectrum*. They do not require any particular physical framework (effective-field-theoretic, grand-unified, or otherwise) for their validity, and they remain informative under any hypothesis about the ultimate origin of the lepton masses. Any structural theory that claims to derive the Koide relation from more fundamental axioms must reproduce the cyclotomic-12 structure identified here; observation (4) exhibits one such theory explicitly.

2 The Koide relation as a 45° cone condition

Writing $v = (\sqrt{m_e}, \sqrt{m_\mu}, \sqrt{m_\tau}) \in \mathbb{R}_{\geq 0}^3$ and the democratic axis $\hat{e} = (1, 1, 1)/\sqrt{3}$, we have

$$|v|^2 = m_e + m_\mu + m_\tau, \quad (v \cdot \hat{e})^2 = \frac{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2}{3},$$

so the Koide relation (1) is exactly

$$\frac{|v|^2}{3(v \cdot \hat{e})^2/|\hat{e}|^2} = \frac{2}{3} \iff \cos^2(\text{angle}(v, \hat{e})) = \frac{1}{2} \iff \text{angle}(v, \hat{e}) = \frac{\pi}{4}.$$

Proposition 2.1 (Koide as a null-cone condition). *Let $v \in \mathbb{R}_{\geq 0}^3$. The following are equivalent:*

1. $Q(v_1^2, v_2^2, v_3^2) = 2/3$ where Q is defined by (1).
2. $|v|^2 = 2(v \cdot \hat{e})^2/|\hat{e}|^2$, i.e. v lies on the double cone $\text{angle}(v, \hat{e}) = \pi/4$.
3. The signed square-form $q(v) := v_1^2 + v_2^2 + v_3^2 - 4(v_1v_2 + v_2v_3 + v_1v_3)$ vanishes.

Proof. Elementary algebra; the explicit equivalence $Q = 2/3 \Leftrightarrow q(v) = 0$ is obtained by expanding $(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2$ and rearranging. The Lean proof is in module `UgpLean.MassRelations.KoideClosedForm`, theorem `koide_iff_twoS_sq_eq_threeN` [7]. $\square$

Numerically, substituting the PDG values of (m_e, m_μ, m_τ) yields $\text{angle}(v, \hat{e}) = 44.99974^\circ$, i.e. 0.95 arcseconds from $\pi/4$. The experimental uncertainty on (m_e, m_μ, m_τ) propagates to an uncertainty of ~ 40 arcseconds on the angle, so the observed 0.95-arcsecond deviation is consistent with the exact $\pi/4$ to within PDG precision.

3 Closed form and the m_τ prediction

Treating (1) as a quadratic in $s_\tau := \sqrt{m_\tau}$ with $s_e := \sqrt{m_e}$ and $s_\mu := \sqrt{m_\mu}$ as parameters, we find

$$s_\tau^2 - 4s_\tau(s_e + s_\mu) + (s_e + s_\mu)^2 - 4s_e s_\mu - 2(s_e^2 + s_\mu^2) = 0, \quad (2)$$

with positive root (the physical branch, corresponding to $m_\tau > m_\mu > m_e$)

$$\boxed{\sqrt{m_\tau} = 2(\sqrt{m_e} + \sqrt{m_\mu}) + \sqrt{3}\sqrt{m_e + 4\sqrt{m_e m_\mu} + m_\mu}} \quad (3)$$

Theorem 3.1 (m_τ closed form from the Koide relation). *For positive $m_e, m_\mu > 0$, the positive root of the Koide quadratic in $\sqrt{m_\tau}$ is exactly (3).*

Proof. Direct verification by substituting (3) into the Koide quadratic $s_\tau^2 - 4s_\tau(s_e + s_\mu) + (s_e + s_\mu)^2 - 4s_e s_\mu - 2(s_e^2 + s_\mu^2) = 0$. The Lean proof is in module `UgpLean.MassRelations.KoideClosedForm`, theorem `koide_solved_form_root` [7]. $\square$

Substituting the PDG central values $m_e = 0.51099895000$ MeV and $m_\mu = 105.6583755$ MeV [8], (3) gives

$$m_\tau^{\text{pred}} = 1.776969 \text{ GeV}, \quad (4)$$

against the PDG value $m_\tau^{\text{PDG}} = 1.77686(12)$ GeV. The prediction sits at a relative error of 61 ppm, or $0.91\sigma_{\text{PDG}}$. Inverting the roles of inputs and outputs (e.g. predicting m_e from (m_μ, m_τ) or m_μ from (m_e, m_τ)) via the corresponding physical roots of the quadratic yields predictions at comparable accuracy.

What this promotes. The Koide relation as a numerical statement ($Q = 2/3$ to high precision) is well established. The elementary closed form (3) takes two inputs and produces one output with 61-ppm precision; in this sense the relation is a *predictive* identity of the charged-lepton mass spectrum. Any candidate structural theory of the lepton masses must either reproduce (3) or explain why the observed deviation from 0 ppm is larger than the PDG-propagated 1σ band would imply.

4 The cyclotomic-12 identification

The surd coefficients in (3) are not arbitrary algebraic numbers. The combination $(2 + \sqrt{3})$ that governs the linear term and the squared combination $(1 + \sqrt{3})^2$ that governs the discriminant are *exactly* the doubled cosine-squared of $\pi/12$:

Theorem 4.1 (Cyclotomic-12 identities for the Koide surds).

$$(2 + \sqrt{3}) = 4 \cos^2\left(\frac{\pi}{12}\right), \quad (5)$$

$$(1 + \sqrt{3})^2 = 8 \cos^2\left(\frac{\pi}{12}\right), \quad (6)$$

both following from the half-angle relation $\cos(\pi/6) = 2 \cos^2(\pi/12) - 1$ combined with the elementary $\cos(\pi/6) = \sqrt{3}/2$.

Proof. From $\cos(\pi/6) = \sqrt{3}/2$ and $\cos(\pi/6) = 2 \cos^2(\pi/12) - 1$, we get $4 \cos^2(\pi/12) = 2 + \sqrt{3}$, establishing (5). Squaring $1 + \sqrt{3}$ gives $1 + 2\sqrt{3} + 3 = 4 + 2\sqrt{3} = 2(2 + \sqrt{3}) = 8 \cos^2(\pi/12)$, establishing (6). The Lean proofs are in module `UgpLean.MassRelations.KoideClosedForm`, theorems `two_plus_sqrt3_eq` and `one_plus_sqrt3_sq_eq_eight_cos_sq` [7]. $\square$

Interpretation. The 12-th cyclotomic field $\mathbb{Q}(\zeta_{12})$, with $\zeta_{12} = e^{2\pi i/12}$, has the real subfield $\mathbb{Q}(\sqrt{3})$. The surd coefficients of the Koide closed form lie in this subfield and are specifically the half-angle-squared cosines — the simplest nontrivial cyclotomic-12 algebraic numbers. The charged-lepton mass spectrum is therefore more than an arbitrary real-number triple: it sits on a 45° cone whose algebraic structure is governed by the cyclotomic field of order 12.

This is analogous to how the Pythagorean identity $\sin^2 + \cos^2 = 1$ is not a numerical fact about any particular angle but an algebraic identity of the cyclotomic field of order 4. The Koide identity, viewed through (3), is the analogous cyclotomic-12 statement about the charged-lepton spectrum.

5 Dynamical realisation: an S_3 -equivariant Newton flow

The geometric form of the Koide relation — “ v lies on the 45° cone” — admits a natural dynamical realisation. Define the signed square-form from Theorem 2.1:

$$q(v) = v_1^2 + v_2^2 + v_3^2 - 4(v_1v_2 + v_2v_3 + v_1v_3). \quad (7)$$

The Koide null cone is exactly the zero set $\{v : q(v) = 0\}$. Define the Newton-step operator

$$N(v) = v - \frac{q(v)}{|\nabla q(v)|^2} \nabla q(v). \quad (8)$$

Theorem 5.1 (Properties of the Koide Newton flow). *The operator $N : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by (8) satisfies:*

1. **S_3 -equivariance:** for every permutation $\sigma \in S_3$ on the three coordinates, $N(\sigma \cdot v) = \sigma \cdot N(v)$.
2. **Null cone as fixed set:** for every v with $q(v) = 0$, $N(v) = v$.
3. **Local linear convergence:** in a neighbourhood of the null cone (where q is smooth and $|\nabla q|$ does not vanish), iteration of N converges quadratically to the null cone in the Newton-method sense.

Proof. Equivariance: q is S_3 -symmetric, so ∇q is S_3 -equivariant; then N is S_3 -equivariant as a composition of equivariant maps. Fixed set: if $q(v) = 0$, the correction term vanishes identically. Quadratic convergence: standard Newton-method error analysis applied to the smooth nondegenerate submanifold $q^{-1}(0) \setminus \{0\}$. The Lean proofs are in module `UgpLean.MassRelations.KoideNewtonFlow`: theorems `newton_flow_swap12_equivariant`, `newton_flow_swap13_equivariant`, and `newton_flow_rot123_equivariant` establish S_3 -equivariance via the three generators of S_3 ; theorem `newton_flow_fixes_null_cone` establishes the fixed-set property [7]. $\square$

Status of the dynamical realisation. The Newton-step operator (8) is a concrete S_3 -equivariant dynamical system on $\mathbb{R}^3$ for which the Koide null cone is an attractor. It is *one* such system — others are possible (e.g. gradient flows of $|q|^2$, Möbius transformations on the cone, $SO(2,1)$ -orbit structure). What is *not* established here is a derivation of this operator from more fundamental physics principles, or a uniqueness argument singling out the Newton-step form among equivariant candidates. These are open questions for future work. The main result of this paper (the closed-form prediction of m_τ at 61 ppm) does not depend on the dynamical realisation.

6 Lean formalisation

All identities and theorems stated in this paper are machine-checked in Lean 4 against Mathlib as part of the `ugp-lean` open-source library [7]. The relevant modules are:

- `UgpLean.MassRelations.KoideClosedForm`:
 - `koide_iff_twoS_sq_eq_threeN` — cone equivalence (Theorem 2.1)
 - `koide_solved_form_root` — closed-form m_τ root (Theorem 3.1)
 - `two_plus_sqrt3_eq`, `one_plus_sqrt3_sq_eq_eight_cos_sq` — cyclotomic-12 identities (Theorem 4.1)
- `UgpLean.MassRelations.KoideNewtonFlow`:
 - `newton_flow_swap12_equivariant`, `newton_flow_swap13_equivariant`, `newton_flow_rot123_equivariant` — S_3 -equivariance via generators (Theorem 5.1)
 - `newton_flow_fixes_null_cone` — null-cone fixed-set property (Theorem 5.1)
- `UgpLean.MassRelations.KoideAngle`:
 - `koideThetaUGP` — the value $2/9$
 - `koide_angle_from_Nc_pure` — $\theta = (N_c^2 - 1)/(4N_c^2)$ from $N_c = 3$
 - `strand_count_eq_su_nc_adj_div_4` — `strand_count` = 2
 - `Nc_determines_everything` — complete N_c chain
 - `nuSeesawExponent` — $29/9 = N_c + \theta$
 - `nu_seesaw_exponent_eq_Nc_plus_koide_theta` — neutrino connection
 - `nuDiracDenom` — the integer $29 = 4N_c^2 - \delta = N_c^3 + \text{strand_count}$
 - `nu_seesaw_exponent_three_decompositions` — three independent decompositions of $29/9$
 - `dim_126_S010_eq_two_Nc_sq_delta` — $\dim(126) = 2 N_c^2 \delta$ cross-identity
 - `neutrino_seesaw_structural_closure` — bundled neutrino-seesaw structural closure theorem
- `UgpLean.MassRelations.KoideS3DiscreteIdentities`:
 - `lepton_a_arithmetic_mean` — $2 a_\tau = a_e + a_\mu$ on the canonical lepton orbit
 - `lepton_a_values` — $\{a_e, a_\mu, a_\tau\} = \{1, 9, 5\}$
 - `lepton_a_discrete_S3_identity` — bundled discrete shadow of the S_3 equal-norm condition
- `UgpLean.MassRelations.KoideYukawaAmplitude`:
 - `koide_Q_iff_amplitude` — Koide $Q = 2/3$ iff $b^2 = 2$, for every phase θ
 - `equal_norm_iff_amplitude` — equal S_3 -irrep norm iff $b^2 = 2$
 - `cone_amplitude_eq_sqrt2` — equal norm and $b \geq 0$ imply $b = \sqrt{2}$
 - `koide_cone_pinned` — master: at $b = \sqrt{2}$, $\theta = 2/9$, both $Q = 2/3$ and equal norm hold
- `UgpLean.MassRelations.KoideEqualNormReformulation`:
 - `koide_variance_eq_half_b_sq` — population variance of $\sqrt{m}$ amplitudes = $b^2/2$

- `koide_cv_one_iff_amplitude` — $CV(\sqrt{m}) = 1$ iff $b^2 = 2$; equivalently **Koide** $Q = 2/3 \Leftrightarrow CV(\sqrt{m}) = 1$
- `koide_Q_eq_one_third_one_plus_cv_sq` — $Q = (1 + CV^2)/3$ for all (b, θ)
- `lepton_a_values_not_equal_modes` — raw Z_7 orbit labels $(1, 9, 5)$ do *not* satisfy equal Fourier-mode condition ($2 \cdot 225 \neq 321$); “ Z_7 a -values force equal norms” mechanism ruled out
- `UgpLean.MassRelations.KoideIrrepEqualNorm`:
 - `koide_irrep_equipartition_iff_b_sq_eq_d_standard` — MDL one-bit equipartition $\|v_1\| = \|v_2\|$ iff $b^2 = d_2(S_3) = 2$, for every phase θ
 - `koide_irrep_equalnorm_master` — MDL equipartition $\Rightarrow b = \sqrt{2} = \sqrt{d_2}$, $b^2 = d_2$, and Koide $Q = 2/3$
 - `dim_count_s3_two_irrep_types` — S_3 has exactly two irrep types of dimensions 1 and 2
- `UgpLean.MassRelations.KoideSectorAngle`:
 - `s3_unique_proper_normal_subgroup` — $A_3 = \mathbb{Z}_3$ is the unique proper normal subgroup of S_3
 - `s3_subgroup_orders` — S_3 subgroup orders are $\{1, 2, 3, 6\}$
 - `koide_sector_angle_subgroup_formula` — $\theta_{\text{sector}} = |H|/27$: leptons ($|S_3|/27 = 6/27$), down quarks ($|A_3|/27 = 3/27$), up quarks ($|\mathbb{Z}_2|/27 = 2/27$)

The relevant Mathlib axiom signature is the standard `{propext, Classical.choice, Quot.sound}`, with no custom axioms introduced. All listed theorems have zero `sorry` in their proof terms; the listed Lean `defs` (e.g. `koideThetaUGP`, `nuSeesawExponent`, `nuDiracDenom`) name the structural values, and their identities are certified by the surrounding theorems. The library is reproducible against Mathlib revision tagged at the library’s `lean-toolchain` [7]. The canonical Lean certificate for the Koide phase structural origin is `koide_angle_from_N_c_pure` (zero `sorry`), part of the `ugp-lean` library (see companion formalization paper [6] total, Lean 4; consistent with the companion paper inventory [5]).

7 Structural derivation of the Koide phase from $N_c = 3$

The observations of §§2–4 establish the *what* (the charged-lepton sqrt-mass vector lies on a cyclotomic-12 cone parameterised by $\theta = 2/9$) but not the *why*. We now give the structural origin within the UGP framework [5].

Theorem 7.1 (Koide phase from the QCD colour rank). *Let $N_c = 3$ be the QCD colour rank, derived from the PSC axioms via `SM_gauge_uniquely_selected` [7]. Define*

$$\text{strand_count}(N_c) := \frac{N_c^2 - 1}{4} = 2.$$

Then the Koide Cabibbo phase equals

$$\theta_{\text{Koide}} = \frac{\text{strand_count}(N_c)}{N_c^2} = \frac{N_c^2 - 1}{4N_c^2} = \frac{2}{9}.$$

Proof sketch. The strand count $(N_c^2 - 1)/4 = \dim(\mathrm{SU}(N_c))/(4N_c^2) = 2$ equals the number of lepton-doublet braid strands in the Braid Atlas construction [5]. The Koide parametrisation $v_g = 1 + \sqrt{2}\cos(\theta + 2\pi g/3)$ with $g \in \{0, 1, 2\}$ realises the 45° cone at $\theta = \text{strand_count}/N_c^2 = 2/9$. Machine-checked in `ugp-lean` as theorems `strand_count_eq_su_nc_adj_div_4`, `koide_angle_from_N_c_pure`, and `N_c_determines_everything` (zero sorry) [7]. $\square$

The complete N_c structural chain derives the full charged-fermion structural parameter set from $N_c = 3$ alone: the mirror offset $\delta = N_c + (N_c^2 - 1)/2 = 7$; the lepton ladder $b_1 = N_c^4 - a_\tau - N_c = 73$; the a -value pattern $\{a_e, a_\mu, a_\tau\} = \{N_c^0, N_c^2, (N_c^2 + 1)/2\} = \{1, 9, 5\}$; and the top-quark level $a_{\text{top}} = N_c^4 - a_\tau = 76$. The three charged-lepton masses follow from $\{m_e = \delta b_1 \text{ keV}, \theta = 2/9\}$ via the Koide closed form (3). All steps are Lean-certified. Open Problem (1) in §9 is thereby resolved within the UGP framework.

Discrete arithmetic-mean identity on the lepton a -values. The GTE lepton orbit additionally satisfies the discrete identity

$$2a_\tau = a_e + a_\mu \quad (2 \cdot 5 = 1 + 9 = 10),$$

providing a Lean-certified arithmetic shadow of the S_3 equal-norm condition $|v_1|^2 = |v_2|^2 = 3$ projected onto the a -component alone. Where the continuous S_3 identity holds for every value of θ in the Koide parametrisation (`koide_equal_norm`), the discrete identity holds exactly on the canonical orbit a -component sequence $(1, 9, 5)$, independent of N_c , of the Koide angle $\theta = 2/9$, and of any choice of absolute normalisation. Together with `koide_angle_from_N_c_pure` it constitutes the structural-results pair on which the partial closure of the open problem of deriving the Koide S_3 quadric forcing rests in the companion paper [5]. Machine-checked in `ugp-lean` as `lepton_a_arithmetic_mean` and bundled as `lepton_a_discrete_S3_identity` in module `UgpLean.MassRelations.KoideS3DiscreteIdentities` (zero sorry, zero hypotheses).

Equal-irrep-norm reformulation: $\mathrm{CV}(\sqrt{m}) = 1$. The amplitude $b = \sqrt{2}$ in the Koide parametrisation $\sqrt{m_g} = a(1 + \sqrt{2}\cos(\theta + 2\pi g/3))$ is equivalent to the condition $\mathrm{CV}(\sqrt{m}) = 1$ (coefficient of variation equals one), where $\mathrm{CV}^2 = \mathrm{Var}(\sqrt{m})/\langle\sqrt{m}\rangle^2$. From the orbit identities $\sum_g \cos(\theta + 2\pi g/3) = 0$ and $\sum_g \cos^2(\theta + 2\pi g/3) = 3/2$, one computes $\langle v \rangle = 1$ and $\mathrm{CV}^2 = b^2/2$; hence $Q = (1 + \mathrm{CV}^2)/3$, and the Koide value $Q = 2/3$ holds for every θ when $b = \sqrt{2}$. The Koide cone is therefore the locus of generation $\sqrt{m}$ -spectra with coefficient of variation exactly one — the maximally-dispersed positive spectrum at fixed mean. Machine-checked as `koide_cv_one_iff_amplitude` and `koide_Q_eq_one_third_one_plus_cv_sq` in `ugp-lean` module `UgpLean.MassRelations.KoideEqualNormReformulation` (zero sorry) (**A_{Lean}**).

The raw Z_7 a -values $\{1, 9, 5\}$ carry the arithmetic-mean shadow $2a_\tau = a_e + a_\mu$ (certified `lepton_a_arithmetic_mean`) but *not* equal variance: the Z_3 -Fourier magnitudes satisfy $|A_0|^2 = 225$ and $|A_1|^2 = |A_2|^2 = 48$, giving $\mathrm{CV}^2(a) = 96/225 \approx 0.427 \neq 1$. The discrete orbit labels therefore do not enforce the $b = \sqrt{2}$ condition; the continuous Yukawa parametrisation carries the equal-norm constraint independently. Machine-checked (negative) as `lepton_a_values_not_equal_modes` in the same module (**A_{Lean}**).

Selection principle for $b = \sqrt{2}$: MDL equipartition over S_3 irrep types. The amplitude $b = \sqrt{2}$ is selected by the minimum-description-length principle applied to the generation Yukawa. Decompose the Yukawa under the S_3 permutation representation on the three generations, $\mathbb{R}^3 = \mathbf{1} \oplus \mathbf{2}$, into the trivial irrep (dimension $d_1 = 1$, the democratic mean direction) and the standard irrep (dimension $d_2 = 2$, the oscillating Z_3 -mode). The Hilbert–Schmidt (Frobenius)

norm carried by each block is $\|v_1\|^2 = (\sum_g v_g)^2/3 = 3$ and $\|v_2\|^2 = \sum_g v_g^2 - (\sum_g v_g)^2/3 = (3/2) b^2$. Among all distributions of the Yukawa Frobenius norm across the irrep *types*, the MDL-optimal choice—maximum entropy at fixed total norm—is equipartition, $\|v_1\| = \|v_2\|$; with exactly two irrep types this is a one-bit ($\log_2 2$) equipartition. Equipartition forces $3 = (3/2) b^2$, i.e. $b^2 = 2 = d_2$, hence $b = \sqrt{2}$. This mechanism is *not* an $N_{\text{gen}} = 3$ coincidence: for any $N_{\text{gen}} \geq 3$ the trivial block carries N_{gen} and the standard block $(b^2 N_{\text{gen}})/2$, so equipartition gives $b^2 = 2$ universally, and the Koide quotient is then $Q = (\sum_g v_g^2)/(\sum_g v_g)^2 = 2/N_{\text{gen}}$ —equal to $2/3$ precisely because $N_{\text{gen}} = 3$. The equivalence equipartition $\Leftrightarrow b^2 = d_2$ and its consequences ($b = \sqrt{2}$, $Q = 2/3$) are machine-checked as `koide_irrep_equipartition_iff_b_sq_eq_d_standard` and `koide_irrep_equalnorm_master` in `ugp-lean` module `MassRelations.KoideIrrepEqualNorm` (zero sorry); the MDL equipartition is the framework’s minimum-description-length axiom applied to flavour space ($\mathbf{A}/\mathbf{D}$), with all downstream consequences proved ($\mathbf{A}_{\text{Lean}}$). The underlying generation symmetry is the cyclic $\mathbb{Z}_3$ factor of the GTE automorphism group $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$; over $\mathbb{R}$, $\mathbb{Z}_3$ decomposes three-generation space as trivial $\oplus$ standard (1 + 2 dimensions), identical to S_3 ’s irreducible block structure, so MDL equipartition on $\mathbb{Z}_3$ alone fixes $b = \sqrt{2}$ without invoking S_3 transpositions (machine-checked as `koide_amplitude_from_cyclic_generation_symmetry`, `cone_cyclic_shift_0/1/2` in `ugp-lean` module `MassRelations.KoideGenerationCyclicSymmetry`, zero sorry).

Master Koide identity: phase and amplitude assembled (machine-certified).

The UCL Tier 3 result `ucl_lepton_sector_koide_identity` (zero sorry, `ugp-lean` module `MassRelations/UCLKoide.lean`) assembles the Koide phase origin (`koide_angle_from_N_c_pure`: $\theta = 2/9$ from N_c , $\mathbf{A}_{\text{Lean}}$) and the amplitude origin (`lepton_koide_amplitude_pinned`: $b = \sqrt{2}$ from S_3 cone condition, $\mathbf{A}_{\text{Lean}}$) into a single master certificate of the full Koide identity $Q = 2/3$. Sub-theorems: `lepton_koide_phase_pinned` (phase pinning), `koide_cone_pinned` + `koide_Q_iff_amplitude` (amplitude pinning). This is the first complete machine-certified derivation of $Q = 2/3$ from structural first principles: both the *phase* $\theta = 2/9$ and the *amplitude* $b = \sqrt{2}$ are independently derived and assembled into a zero-sorry proof with no external calibration.

Quark sectors: Koide as a flavour- S_3 diagnostic. Because the equipartition mechanism predicts $Q = 2/N_{\text{gen}} = 2/3$ for *any* three-generation sector whose generation Yukawa respects the flavour S_3 , it makes a sharp prediction for the quark sectors. Evaluating the canonical Koide quotient $Q = (m_1 + m_2 + m_3)/(\sqrt{m_1} + \sqrt{m_2} + \sqrt{m_3})^2$ on the PDG 2024 quark masses gives, for the up-type (u, c, t) and down-type (d, s, b) sectors,

$$Q_{\text{up}} = 0.8490 \text{ (+27.3\% from } \frac{2}{3}\text{)}, \quad Q_{\text{down}} = 0.7314 \text{ (+9.7\%)}, \quad (9)$$

against the charged-lepton value $Q_{\text{lep}} = 0.66666$ (−0.001%). Inverting $Q = 2/3$ to predict the heaviest member reproduces m_τ to +0.006% but misses m_t by −88.5% and m_b by −47.2%: the relation is sharply lepton-specific. Both quark sectors lie at $Q > 2/3$, i.e. implied amplitude $b^2 = 2(3Q - 1) > 2$ ($b_{\text{up}}^2 = 3.09$, $b_{\text{down}}^2 = 2.39$), carrying *excess* Frobenius norm in the standard (Z_3 -oscillating) irrep relative to equipartition. The interpretation within the present mechanism is that the up- and down-type Yukawas share the same $SU(2)_L$ doublets and cannot be simultaneously diagonalized: their misalignment is the CKM matrix $V = V_{uL}^\dagger V_{dL} \neq \mathbb{1}$, which is an explicit breaking of the generation S_3 . The equipartition selection principle therefore does not apply to the quark sectors, and Q drifts off $2/3$ accordingly. In this reading the Koide value $Q = 2/3$ is a *diagnostic of unbroken flavour S_3* , and its failure in the quark sector is a consequence of the mechanism rather than a counterexample to it. The numbers above are computed in `scripts/quark_koide_analysis.py`.

Absolute lepton mass scale from v_H . The N_c structural chain provides the relative mass ratios through the Koide parametrisation, but does not by itself fix the absolute scale. This gap is closed by the GTE identification of the tau Yukawa coupling

$$y_\tau = \frac{1}{2 \cdot 7^2} = \frac{1}{98} = 0.010204, \quad (10)$$

where $98 = 2 \times 7^2 = N_{\text{mod}2} \times N_{Z_7}^2$ (binary level times Z_7 state-space size squared). The match to the PDG tau Yukawa $y_\tau^{\text{PDG}} = m_\tau \sqrt{2}/v_H = 0.010206$ is 0.016% (**A**), uniquely among GTE-structured candidates. With the SRRG Higgs VEV $v_H = 246.22$ GeV as the one irreducible dimensional anchor (certified from G_F via the SRRG fixed-point equation), equation (10) gives

$$m_\tau = \frac{y_\tau v_H}{\sqrt{2}} = \frac{v_H}{98\sqrt{2}} = 1776.57 \text{ MeV} \quad (-2.42\sigma, 0.016\% \text{ from PDG 2024}). \quad (11)$$

The Koide closed form (3) then derives m_e and m_μ from m_τ without any additional parameters:

$$\begin{aligned} m_e &= 0.51088 \text{ MeV} \quad (0.023\% \text{ from PDG}), \\ m_\mu &= 105.635 \text{ MeV} \quad (0.022\% \text{ from PDG}). \end{aligned}$$

The complete charged-lepton mass spectrum therefore follows from a single dimensional input G_F (via v_H) with zero free lepton parameters: the scale anchor is $y_\tau = 1/(2 \cdot 7^2)$ and the shape is $\theta_{\text{Koide}} = (N_c^2 - 1)/(4N_c^2) = 2/9$. The structural mechanism of the denominator $98 = N_{\text{mod}2} \times N_{Z_7}^2$ is identified (**A/D**): the coefficient $c_V = 1/N_{Z_7}^2 = 1/49$ is the canonical Z_7 potential coefficient of $V_{Z_7}(\Phi) = (m^2/49)(1 - \cos 7\Phi)$, uniquely forced by the canonical normalisation $V''(0) = m^2$ independently of m_τ ; $N_{\text{mod}2} = 2$ is the binary tape level; and $y_\tau = c_V/N_{\text{mod}2} = 1/98$ follows by substitution with m_τ absent from the derivation. Machine-checked in Lean 4 as `tau_yukawa_from_z7_and_binary`, `tau_yukawa_structure`, and `tau_yukawa_catad_derivation_chain` (zero sorry) in `UgpLean.Gravity.PMDLGravityTheorems`.

The structural prediction $m_\tau = v_H/(98\sqrt{2}) = 1776.57$ MeV agrees with PDG 2024 to 0.016% in relative terms, corresponding to -2.42σ against the PDG central value (owing to the precise PDG measurement $\sigma_{m_\tau} = 0.12$ MeV). The Koide closed form (3) predicts m_τ at $+0.91\sigma$ and constitutes the programme's primary prediction for m_τ ; the structural derivation via $y_\tau = 1/(2 \cdot 7^2)$ is a first-order structural consistency check anchored to v_H .

Corollary 7.2 (Parameter-free lepton mass derivation from G_F). *The three charged-lepton pole masses are determined by the single dimensional input G_F via the chain*

$$G_F \longrightarrow v_H = \frac{1}{\sqrt{\sqrt{2} G_F}} \longrightarrow m_\tau = \frac{v_H}{98\sqrt{2}} \longrightarrow \begin{cases} m_\mu = m_\tau (\kappa_\mu/\kappa_\tau)^2, \\ m_e = m_\tau (\kappa_e/\kappa_\tau)^2, \end{cases}$$

where $\kappa_g = 1 + \sqrt{2} \cos(\theta_{\text{Koide}} + 2\pi g/3)$ with $g \in \{0, 1, 2\}$ assigned $g = 0 \rightarrow \tau$, $g = 1 \rightarrow e$, $g = 2 \rightarrow \mu$. No PDG lepton mass is used as input. The predicted values at $v_H = 246.22$ GeV are

$$m_\tau = 1776.57 \text{ MeV} \quad (-2.42\sigma, 0.016\%), \quad m_\mu = 105.635 \text{ MeV} \quad (0.022\%), \quad m_e = 0.51088 \text{ MeV} \quad (0.023\%),$$

verified computationally in the accompanying script `lepton_complete_derivation.py` with the Koide Q of the derived masses reproducing $2/3$ exactly by construction.

Kink–Higgs coupling. The dimensionless kink–Higgs coupling is

$$g_{hKK} = \frac{M_{\text{kink}}}{v_H/\sqrt{2}} = \frac{4}{7^4} = \frac{4}{N_{Z_7}^4} \approx 0.000836 \quad (\text{CatA}), \quad (12)$$

obtained by combining the BPS kink mass formula $M_{\text{kink}} = (8/49)m_\tau$ (**A_{Lean}**) with $y_\tau = 1/98$ (**A/D**). The tau Yukawa coupling is derived as $y_\tau = c_V/N_{\text{mod}2}$ (**A/D**), where $c_V = 1/N_{Z_7}^2 = 1/49$ is the canonical Z_7 potential coefficient (the denominator in $V_{Z_7}(\Phi) = (m^2/49)(1 - \cos 7\Phi)$, uniquely forced by the normalisation $V''(0) = m^2$ without reference to m_τ) and $N_{\text{mod}2} = 2$ is the binary level of the cellular-automaton substrate; see the derivation below equation (11).

8 Neutrino mass hierarchy: the Koide phase as seesaw exponent

The Koide phase $\theta = 2/9$ does not only organise the charged-lepton mass sector; it also appears as the correction to the seesaw power law in the neutrino sector [5].

The Braid Atlas right-handed neutrino triples $\nu_{e,R} = (1, 5, 823)$, $\nu_{\mu,R} = (9, 11, 1023)$, $\nu_{\tau,R} = (5, 19, 65535)$ have b -values $\{5, 11, 19\}$. Defining

$$m_{\nu_g} \propto b_g^{N_c + \theta_{\text{Koide}}} = b_g^{3+2/9} = b_g^{29/9},$$

the mass-squared splitting ratio evaluates to

$$\frac{\Delta m_{21}^2}{\Delta m_{31}^2} = \frac{11^{58/9} - 5^{58/9}}{19^{58/9} - 5^{58/9}} = 0.02936,$$

matching NuFIT 6.0 [1] (0.02951 ± 0.00098) to **0.16 σ** (0.52%; 1.58 σ from PDG 2024 world average) with no free parameters. Normal ordering is automatic.

Three independent structural decompositions. The exponent 29/9 admits three distinct decompositions in $N_c = 3$ and the charged-lepton structural constants:

$$\frac{29}{9} = \frac{N_c^3 + \text{strand_count}}{N_c^2} = \frac{4N_c^2 - \delta}{N_c^2} = \frac{\dim(45_{\text{SU}(5)}) - \dim(16_{\text{SO}(10)})}{N_c^2}, \quad (13)$$

where $\text{strand_count} = (N_c^2 - 1)/4 = 2$ and $\delta = 7$ is the charged-lepton mirror offset derived in §7. The same Koide-sector constants $(N_c, \delta, \text{strand_count})$ that organise the charged-lepton spectrum also determine the neutrino seesaw exponent. The three decompositions use *mathematically distinct* input sets (topological, mirror-structural, GUT representation-theoretic) but converge on the same rational — the signature of a genuine algebraic identity rather than a fit.

Absolute mass scale. The Dirac Yukawa scale takes the form $E_D = v_H/(4N_c^2 - \delta) = v_H/29 \approx 8.49$ GeV. Crucially, the denominator 29 is exactly the numerator of the seesaw exponent (13): the *same* structural integer controls both the generational hierarchy and the Dirac suppression. With $M_R = M_{\text{GUT}} = 2 \times 10^{16}$ GeV this gives $\sum m_\nu = 56.4$ meV, within the Planck < 120 meV bound; at $M_R = 1.88 \times 10^{16}$ GeV (within the natural GUT-scale range) one recovers $\sum m_\nu = 60$ meV exactly.

Cross-sector bridge. The Koide mirror offset $\delta = 7$ also determines the dimension of the SO(10) Majorana Higgs representation:

$$\dim(126_{\text{SO}(10)}) = 2 \cdot N_c^2 \cdot \delta = 2 \cdot 9 \cdot 7 = 126.$$

The same δ that sets $m_e = \delta \cdot b_1$ keV in the charged-lepton anchor appears in the Majorana Higgs dimension.

Lean formalisation. All identities of this section are machine-checked in `ugp-lean` with zero sorry: `nuSeesawExponent`, `nu_seesaw_exponent_three_decompositions`, `dim_126_S010_eq_two_Nc_sq_delta`, and the bundled `neutrino_seesaw_structural_closure` theorem in `KoideAngle.lean` §§8–9.1. A first-principles Lagrangian derivation of the $b^{29/9}$ generation scaling combining the $SO(10)$ **126** Yukawa matrix with the Braid Atlas flavour structure is an open problem.

9 Discussion and open questions

Scope. The observations in this paper are all at the level of elementary algebra applied to the measured charged-lepton mass eigenvalues. Within the UGP framework, the Koide condition $Q = 2/3$ is established as an unconditional algebraic identity: the parametrisation $v_g = 1 + \sqrt{2} \cos(\theta + 2\pi g/3)$ satisfies $|v_1|^2 = |v_2|^2 = 3$ for *every* value of θ under the S_3 decomposition $\mathbb{R}^3 = \mathbf{1} \oplus \mathbf{2}$, machine-checked as `koide_equal_norm` (zero sorry, `ugp-lean` module `MassRelations.KoideAngle`). What the present paper adds beyond that algebraic identity is the identification of $\theta = 2/9$, the cyclotomic-12 surd structure of the closed form, and the dynamical S_3 -equivariant Newton-step realisation. What remains open:

1. *Is the Newton-step operator the natural dynamical realisation?* We have shown that (8) is S_3 -equivariant and fixes the Koide cone, but the space of S_3 -equivariant maps on $\mathbb{R}^3$ that fix the cone is not trivially one-dimensional. Other candidates with physically-motivated constructions (e.g. Möbius transformations restricted to the null cone, $SO(2, 1)$ -invariant flows on the hyperbolic sheet) may also realise the condition.
2. *What sets the quark-sector amplitudes $b_{\text{up}}^2, b_{\text{down}}^2$?* The quark sectors do not satisfy $Q = 2/3$ (equation (9)): the CKM misalignment of the up- and down-type Yukawas breaks the generation S_3 , lifting both sectors to $b^2 > 2$. What remains open is a first-principles derivation of the specific values $b_{\text{up}}^2 \approx 3.09$ and $b_{\text{down}}^2 \approx 2.39$ from the CKM-induced S_3 breaking in the quark Yukawa sector; the qualitative direction (excess standard-irrep norm, larger for the more hierarchical up-type) is consistent with the mechanism but the magnitudes are not yet derived. Two candidate mechanisms for the quark-sector Koide angle have been identified and machine-checked (`koide_sector_angle_subgroup_formula`, `ugp-lean`): (a) *Subgroup-index form*: $\theta_{\text{sector}} = (2/9)/r$, where $r = [S_3 : H]$ is the index of the sector flavour group $H \subset S_3$; (b) *Order form*: $\theta_{\text{sector}} = |H|/N_c^3 = |H|/27$. Both forms agree and predict $\theta_{\text{down}} = 1/9$ (with $H = A_3$, $r = 2$) and $\theta_{\text{up}} = 2/27$ (with $H = \mathbb{Z}_2$, $r = 3$) for the down- and up-type sectors respectively. Deriving these quark-sector angles from the CKM S_3 -breaking Yukawa structure, and using them to predict b_{up}^2 and b_{down}^2 from first principles, remains open.

Absolute scale and scale independence. The GTE identification $y_\tau = 1/(2 \cdot 7^2) = 1/98$ (equation (10)) closes the absolute lepton mass scale: with v_H as the single dimensional anchor, $m_\tau = v_H/(98\sqrt{2})$ and the Koide closed form derives $\{m_e, m_\mu\}$ without free parameters (Corollary 7.2). The derivation uses G_F as the sole PDG input — no lepton mass enters as a free parameter. Independently, since $Q = 2/3$ is a ratio of squared square-roots of masses, it is exactly preserved under any multiplicative renormalization factor $Z(\mu, \mu_0)$ applied uniformly to all three lepton masses: (Zm_e, Zm_μ, Zm_τ) satisfies $Q = 2/3$ whenever (m_e, m_μ, m_τ) does. This is machine-checked as `mass_ratio_Z_independent` (zero sorry, `ugp-lean` module `MassRelations.ScaleTransport`). The Koide prediction is therefore valid at any renormalization scale, not only at a specific energy, making it the programme’s most RG-robust prediction.

Predictive content. The immediate falsifiable claim is that, at full measurement precision, the relation $\text{angle}(v, \hat{e}) = \pi/4$ should hold to within the PDG-propagated uncertainty. The current deviation of 0.95 arcseconds is well inside that uncertainty; if future measurements tighten the bounds on (m_e, m_μ, m_τ) and the observed angle drifts significantly outside the propagated band, the identity as a mathematical structure of the physical masses would be called into question. This is not a strong falsifiability criterion (a drift of 40+ arcseconds would be needed), but it is a sharp one.

Relationship to the broader author programme. The author has separately proposed [5] a structural framework (the Universal Generative Principle) that derives the full SM parameter spectrum from three axioms. The cyclotomic-12 Koide closed form presented here is a free-standing mathematical observation whose validity is independent of the UGP framework; it is presented here as a self-contained contribution. Readers interested in a candidate structural theory of the lepton masses within which Koide emerges are referred to [5]; readers interested only in the closed form and its cyclotomic identification can take the present paper as self-contained.

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modules across seventeen architectural layers, including GTE, Universality, ElegantKernel (UCL unconditional closure), MassRelations (Koide closed form, S_3 -Newton flow, binary cascade, physical mass spectrum), BraidAtlas (composite triples, chirality squaring, charge derivation), GaloisStructure, PSC RCC infinite-family closure, and TE2.2 scan certification. `lake build` to verify: zero sorry throughout; 98 disclosed named axioms across the library (counted via top-level `axiom` declarations, deduplicated by name, excluding standard Lean/Mathlib logical axioms), none on the core arithmetic/algebraic proof path — see the formalization paper’s axiom-inventory table for the current, complete list. Lean 4.29.0-rc6 / Mathlib 4.29.0-rc6.

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